Fiscal Year:	FY 2012	**Task Last Updated:**	FY 04/01/2013
PI Name:	Ferl, Robert Ph.D.		
Project Title:	Transgenic Plant Biomonitors of Space Flight Exposure		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:	**TechPort:**		No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Plant Biology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Gainesville	**State:**	FL
Zip Code:	32611-0690	**Congressional District:**	3
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	98-HEDS-02
Start Date:	03/15/2007	**End Date:**	05/31/2012
No. of Post Docs:	2	**No. of PhD Degrees:**	1
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA KSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:	Shuttle/ISS		
Flight Assignment:	STS-129; STS-130 ; STS-131 ; ISS NOTE: New end date is 5/31/2012, per PI (Ed., 2/3/2011) NOTE: Extended to 5/31/2011 per NSSC info; original end date 3/14/2010 (9/2009)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Paul, Anna-Lisa Ph.D. (Co-PI: University of Florida)		
Grant/Contract No.:	NNX07AH27G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	This proposal is designed to be the next-generation of our previous flight project, PGIM-01, and investigates the use of reporter gene technology for the molecular analysis of development and stress in plants within space flight environments. We have designed a series of transgenic plants that carry sensor promoter - reporter gene constructs that are capable of monitoring a variety of environmental and developmental influences that might be altered in microgravity and general space flight environments. The first generation of these experiments with the GUS reporter have been tested for biological veracity on KC-135 parabolic flight missions, and flew as PGIM-01 in the Plant Growth Facility on STS-93. Results of these KC-135 missions and STS-93 were used in this proposal to (1) design a second generation of sensor promoters to refine our definition of the cellular biological impact of microgravity and space flight; (2) design a second generation of reporter gene constructs that include real-time and non- destructive observations of reporter gene activity; (3) design and conduct short and longer term flight experiments that begin a comprehensive analysis of molecular genetic changes induced by space flight; and (4) develop methods to integrate reporter gene activities with data from DNA chip analysis of genome expression. The long-term goal is an understanding of the space flight induced molecular changes in plant gene activity, an understanding that will lead to hardware and/or genetic engineering modifications that will mitigate the negative impacts of space flight on plant growth and development.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	An understanding of signal transduction in higher plants (and by inference, all eukaryotes) is central to basic molecular biology research.
Task Progress:	The APEX-TAGES investigation was focused on the robust discovery of changes in the patterns of gene expression in Arabidopsis in response to the spaceflight environment. The study was designed with three replicated experiments that supported strong statistical analysis and affirmation that the effects displayed by the plants were true characteristics of spaceflight adaptation, independent of localized environmental phenomenon. Experimental subsets were launched on STS-129, 130, and 131 in 2009/2010. Two approaches were used; GFP reporter genes were used to collect gene expression data in real time within GIS (GFP Imaging System) imaging hardware, growth morphology images were also collected, and plants were harvested on orbit to RNAlater for subsequent gene expression profiling with Affymetrix and RNASeq transcriptome analyses. Leaves, hypocotyls, and roots were examined independently, revealing that there are a number of organ-specific responses that Arabidopsis employs to respond to the spaceflight environment. Almost a third of the statistically significant, differentially expressed genes encode proteins typically associated with cell wall remodeling, including touch, pathogen response and wounding. Many were dramatically induced (between 5 and 25-fold). In addition, the unique imaging capabilities of the GIS housed in the ABRS unit enabled us to collect morphometric data on the patterns of root growth in the GFP reporter gene plants. These data revealed that there are fundamental differences in root growth and development on orbit, and that two common cultivars of Arabidopsis, Wassilewskija (WS) and Columbia (Col-0), are distinct in their responses to spaceflight. The dominance of touch-wound-remodeling responses in the molecular data, combined with the dramatic changes in root growth morphology, lead directly to hypotheses suitable for future proposals.
Bibliography Type:	Description: (Last Updated: 11/27/2022)
Abstracts for Journals and Proceedings	Paul A-L, Ferl RJ. "Root Skewing and Waving Is not Dependent on Gravity in Arabidopsis." 28th Annual Meeting of the American Society for Gravitational and Space Research, New Orleans, LA, November 27-December 2, 2012. 28th Annual Meeting of the American Society for Gravitational and Space Research, New Orleans, LA, November 27-December 2, 2012. Program and abstracts, p. 63. See https://asgsr.org/images/stories/pdf/2012_AnnualMeeting_Program.pdf ; accessed 12/8/14. , Nov-2012
Abstracts for Journals and Proceedings	Ferl RJ, Paul A-L. "Spaceflight Biology in the Suborbital Realm." 28th Annual Meeting of the American Society for Gravitational and Space Research, New Orleans, LA, November 27-December 2, 2012. 28th Annual Meeting of the American Society for Gravitational and Space Research, New Orleans, LA, November 27-December 2, 2012. Program and abstracts, p. 63. See https://asgsr.org/images/stories/pdf/2012_AnnualMeeting_Program.pdf ; accessed 12/8/14. , Nov-2012
Articles in Peer-reviewed Journals	Paul A-L, Denison FC, Schultz ER, Zupanska AK, Ferl RJ. "14-3-3 phosphoprotein interaction networks - does isoform diversity present functional interaction specification?" Frontiers in Plant Science. 2012;3:190. Epub 2012 Aug 20. http://dx.doi.org/10.3389/fpls.2012.00190 ; PubMed PMID: 22934100 , Aug-2012
Articles in Peer-reviewed Journals	Paul A-L, Amalfitano CE, Ferl RJ. "Plant growth strategies are remodeled by spaceflight." BMC Plant Biol. 2012 Dec 7;12:232. http://dx.doi.org/10.1186/1471-2229-12-232 ; PubMed PMID: 23217113 , Dec-2012
Articles in Peer-reviewed Journals	Mayfield JD, Paul A-L, Ferl RJ. "The 14-3-3 proteins of Arabidopsis regulate root growth and chloroplast development as components of the photosensory system." Journal of Experimental Botany. 2012 May;63(8):3061-70. Epub 2012 Feb 29. http://dx.doi.org/10.1093/jxb/ers022 ; PubMed PMID: 22378945 , May-2012
Articles in Peer-reviewed Journals	Paul A-L, Zupanska AK, Ostrow DT, Zhang Y, Sun Y, Li JL, Shanker S, Farmerie WG, Amalfitano CE, Ferl RJ. "Spaceflight transcriptomes: unique responses to a novel environment." Astrobiology. 2012 Jan;12(1):40-56. http://dx.doi.org/10.1089/ast.2011.0696 ; PMID: 22221117 , Jan-2012
Articles in Peer-reviewed Journals	Zupanska AK, Denison FC, Ferl RJ, Paul A-L. "Spaceflight engages heat shock protein and other molecular chaperone genes in tissue culture cells of Arabidopsis thaliana." American Journal of Botany. 2013 Jan;100(1):235-48. http://dx.doi.org/10.3732/ajb.1200343 ; PubMed PMID: 23258370 , Jan-2013
Articles in Peer-reviewed Journals	Paul A-L, Wheeler RM, Levine HG, Ferl RJ. "Fundamental plant biology enabled by the space shuttle." American Journal of Botany. 2013 Jan;100(1):226-34. http://dx.doi.org/10.3732/ajb.1200338 ; PubMed PMID: 23281389 , Jan-2013

Articles in Peer-reviewed Journals	Parsons-Wingerter P, Vickerman MB, Paul A-L, Ferl RJ. "Mapping by VESGEN of leaf venation patterning in Arabidopsis thaliana with bioinformatic dimensions of gene expression." Gravit Space Res. 2014 Aug;2(1):68-81. http://gravitationalandspacebiology.org/index.php/journal/article/view/618 , Aug-2014
Articles in Peer-reviewed Journals	Ferl RJ, Koh J, Denison F, Paul A-L. "Spaceflight induces specific alterations in the proteomes of Arabidopsis." Astrobiology. 2015 Jan;15(1):32-56. Epub 2014 Dec 17. http://dx.doi.org/10.1089/ast.2014.1210 ; PubMed PMID: 25517942; PubMed Central PMCID: PMC4290804 , Jan-2015
Articles in Peer-reviewed Journals	Schultz ER, Zupanska AK, Sng NJ, Paul A-L, Ferl RJ. "Skewing in Arabidopsis roots involves disparate environmental signaling pathways." BMC Plant Biol. 2017 Feb 1;17(1):31. https://doi.org/10.1186/s12870-017-0975-9 ; PubMed PMID: 28143395; PubMed Central PMCID: PMC5286820 , Feb-2017
Articles in Peer-reviewed Journals	Schultz ER, Paul A-L, Ferl RJ. "Root growth patterns and morphometric change based on the growth media." Microgravity Science and Technology. 2016 Dec;28(6):621-31. http://dx.doi.org/10.1007/s12217-016-9514-9 , Dec-2016
Awards	Paul A-L, Amalfitano CE, Ferl RJ. "Runner-Up in Plant biology, Environmental biology and Ecology' category Research Award as judged by leading experts in the field. To be presented presented Sunday, April 21, 2013 at the Experimental Biology FASEB meeting in Boston, USA. " Apr-2013
Journal/Magazine covers	Ferl RJ, Koh J, Denison F, Paul A-L. "Cover in journal Astrobiology for article, 'Spaceflight induces specific alterations in the proteomes of Arabidopsis.' " Astrobiology. 2015 Jan;15(1):32-56. http://dx.doi.org/10.1089/ast.2014.1210 ; PubMed PMID: 25517942 , Jan-2015
Significant Media Coverage	Paul A-L, Amalfitano CE, Ferl RJ. "Article, 'Plant growth strategies are remodeled by spaceflight,' originally published in BMC Plant Biology 12:232 was released to the press and subsequently highlighted in various media." Specific publications are cited in this Task Book Bibliography., Dec-2012
Significant Media Coverage	Perkins S. "ScienceShot: Plants ... in ... Space! Highlights the article by Paul A-L, Amalfitano CE, Ferl RJ, 'Plant growth strategies are remodeled by spaceflight,' in BMC Plant Biol. 2012 Dec 7;12:232." ScienceNow, December 6, 2012. http://news.sciencemag.org/sciencenow/2012/12/scienceshot-plants-in-space.html?ref=hp , Dec-2012
Significant Media Coverage	Science 2.0 News Staff. "Plant Growth in Zero Gravity. Highlights the article by Paul AL, Amalfitano CE, Ferl RJ, 'Plant growth strategies are remodeled by spaceflight,' in BMC Plant Biol. 2012 Dec 7;12:232." Science 2.0 Science Blog. http://www.science20.com/news_articles/plant_growth_zero_gravity-98388 , Dec-2012
Significant Media Coverage	Owen J. "Plants Grow Fine Without Gravity. Highlights the article by Paul AL, Amalfitano CE, Ferl RJ, 'Plant growth strategies are remodeled by spaceflight,' in BMC Plant Biol. 2012 Dec 7;12:232." National Geographic News. Published December 7, 2012 http://news.nationalgeographic.com/news/2012/121207-plants-grow-space-station-science/ , Dec-2012
Significant Media Coverage	Draxler B. "Plants in Space Prove Gravity Unnecessary For Normal Growth. Highlights the article by Paul AL, Amalfitano CE, Ferl RJ, 'Plant growth strategies are remodeled by spaceflight,' in BMC Plant Biol. 2012 Dec 7;12:232." Discover blog. December 12, 2012. http://blogs.discovermagazine.com/80beats/?p=41754&utm_source=dlvr.it&utm_medium=twitter# , Dec-2012
Significant Media Coverage	Wells RH. "Plants in space: Gravity not needed to orient root growth, UF researchers find. Highlights the article by Paul AL, Amalfitano CE, Ferl RJ, 'Plant growth strategies are remodeled by spaceflight,' in BMC Plant Biol. 2012 Dec 7;12:232." University of Florida News. December 12, 2012. http://news.ufl.edu/2012/12/12/gravity-root/ , Dec-2012